# Coimisiún na Scrúduithe Stáit
# State Examinations Commission

## LEAVING CERTIFICATE EXAMINATION, 2022

## PHYSICS AND CHEMISTRY – HIGHER LEVEL

WEDNESDAY, 22 JUNE – MORNING, 9:30 to 12:30

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**Five** questions to be answered.

Answer any **two** questions from Section I, any **two** questions from Section II and **one** other question from either Section I or Section II.

All the questions carry equal marks.

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**N.B.** Relevant data are listed in the *Formulae and Tables* booklet, which is available from the superintendent. Take $g = 9.8$ m s$^{-2}$ as the acceleration due to gravity at the surface of the Earth.

Do not hand this up.

This document will not be returned to the State Examinations Commission.

The 2022 examination papers were adjusted to compensate for disruptions to learning due to COVID-19. This examination paper does not necessarily reflect the same structure and format as the examination papers of past or subsequent years.

# SECTION I – PHYSICS

**1.** Answer **eleven** of the following items (*a*), (*b*), (*c*), etc. All the items carry equal marks. *Keep your answers short*.

(*a*) Is displacement an example of a vector? Give a reason for your answer.

(*b*) In April 2021 the tiny Mars helicopter Ingenuity, shown in **Figure 1**, became the first aircraft to make a powered controlled flight on a planet other than Earth. The mass of Ingenuity on Earth is 1.80 kg. The acceleration due to gravity on the surface of Mars is 3.72 m $s^{-2}$. What is
(*i*) the mass,
(*ii*) the weight, of Ingenuity on the surface of Mars?



**Figure 1**

(*c*) What is a force?

(*d*) Two satellites are in stable orbit around the Earth at the same distance from its centre but the Earth's gravitational force on one satellite is twice that on the other. Explain.

(*e*) What is meant by the *triple point* temperature of water?

(*f*) The resistance of a piece of wire is 110.0 Ω at 0 °C and 150.0 Ω at 100 °C. Calculate the temperature in °C when the resistance of the wire is 132.8 Ω.

(*g*) State Boyle's law.

(*h*) **Figure 2** represents plane water waves approaching a gap. State and explain what happens, if anything, to the frequency of the waves as they pass through the gap.

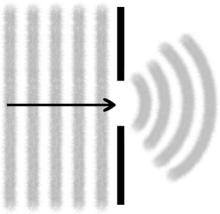


**Figure 2**

(*i*) What type of electromagnetic radiation can be detected
(*i*) using a blackened thermometer bulb,
(*ii*) using fluorescent material?

(*j*) Draw a ray diagram to show the formation of an image in a convex mirror.

(*k*) Why does dispersion occur when a beam of white light passes through a prism?

(*l*) State Coulomb's law of force between two charges.

(*m*) Identify two factors that affect the capacitance of a parallel-plate capacitor.

(*n*) Four 20 μF capacitors are available. Draw diagrams to show how some or all the capacitors could be connected to produce a combined capacitance of (*i*) 60 μF, (*ii*) 5 μF.

(*o*) How can a moving-coil galvanometer be modified for use as a voltmeter?

(*p*) A current carrying wire lying on a table between two bar magnets of equal strength as shown in **Figure 3** experiences an upwards force. Is **A** a north magnetic pole or a south pole?

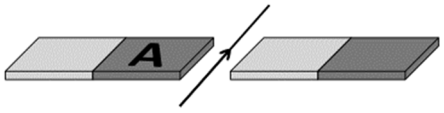



**Figure 3**

(*q*) What is meant by nuclear fusion?

(*r*) The energy released in a nuclear reaction is 1.8 x $10^{-12}$ J. Calculate the loss in mass in kg.

(11 × 6)

**2.** (*a*) State the principle of conservation of energy.
Distinguish between kinetic energy and potential energy.
Define the S.I. unit of energy, the joule. (18)

(*b*) State Newton's first law of motion. (6)

(*c*) **Figure 4** shows a vertical cross section through part of a skateboard park. A small ball of mass 0.40 kg, with an initial speed *u* parallel to **AB** at the point **A**, moved from **A** to **E** staying in contact with the skating surface throughout and without losing any energy due to friction.

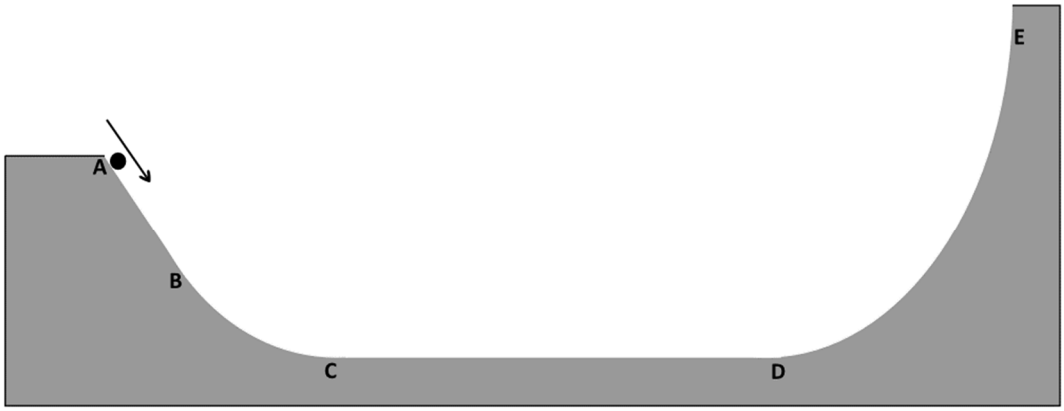



**Figure 4**

The ball travelled 1.25 m from **A** to **B** with constant acceleration of 8.0 m $s^{-2}$ in 0.25 s. Find

(*i*) the initial speed *u* of the ball at **A**,

(*ii*) the speed of the ball at **B**,

(*iii*) the kinetic energy of the ball at **B**.

**B** is 0.50 m above **C**.

(*iv*) Use the principle of conservation of energy to calculate the potential energy converted to kinetic energy as the ball rolled from **B** to **C**.

(*v*) What was the total kinetic energy of the ball when it reached **C**? (24)

The ball had a constant horizontal speed as it moved from **C** to **D**.

(*vi*) Is there a horizontal force on the ball between **C** and **D**?
Explain your answer.

(*vii*) Is there a vertical force on the ball between **C** and **D**?
Explain your answer.

The ball came to rest for an instant at **E** where it reached its maximum height.

(*viii*) Calculate the height of **E** above **D**. (18)

**3.** (*a*) (*i*) State the laws of refraction of light.

(*ii*) State the two conditions necessary for total internal reflection of light to occur.

(*iii*) What is meant by the critical angle of a material?

(*iv*) Sketch a diagram to show how a beam of laser light can travel along an optic fibre by means of total internal reflection. (24)

(*b*) The table gives the values of the critical angle (*c*) for a number of solid materials each of which has a different refractive index (*n*).

| *n* | 1.1 | 1.3 | 1.5 | 1.7 | 2.2 | 2.4 | 2.7 |
|---|---|---|---|---|---|---|---|
| *c* (°) | 65 | 50 | 42 | 36 | 27 | 25 | 22 |

(*i*) Plot a graph of *n* versus $\frac{1}{\sin c}$ to show that *n* and $\sin c$ are inversely proportional.

(*ii*) Use your graph to find the value of *k* in the equation $n = k\frac{1}{\sin c}$.

(*iii*) What is the refractive index of a gemstone that has a critical angle of 30°? (30)

(*c*) When an object of height 5.0 cm was placed in front of a converging (convex) lens a real inverted image of height 2.5 cm was formed on the other side of the lens. The distance between the object and its image was 45 cm.

(*i*) How far from the lens was the object placed?

(*ii*) Calculate the focal length of the lens. (12)

**4.** (*a*) (*i*) What is meant by the term *ideal gas*?

(ii) State two assumptions of the kinetic theory of gases.

(iii) What property of a gas relates to the average kinetic energy of its particles? (15)

**5.** Under a microscope it was observed that tiny pollen grains suspended in water moved in an irregular zig-zag manner.

(i) What name is given to this irregular movement?

(ii) What is the explanation for the irregular movement?

(*iii*) What is the significance of this observation? (12)

(*c*) The data in the table below shows the product of the pressure and the volume (*pV*) of a fixed mass of **$H_2$** gas at a number of different absolute temperature (*T*) values.

| *T* (K) | 100 | 150 | 200 | 250 | 300 | 350 | 400 |
|---|---|---|---|---|---|---|---|
| *pV* (J) | 29.1 | 44.0 | 58.2 | 72.7 | 86.4 | 101.8 | 116.3 |

(*i*) Justify why *pV* can be measured in joules.

(*ii*) It is not possible to get a *pV* measurement for hydrogen at 0 K. Explain. (12)

(*iii*) Plot a graph of *pV* versus *T* for the **$H_2$** gas.

(*iv*) Use the slope of your graph to find the number of moles of **$H_2$** gas present.

(*v*) How would the slope of the graph be affected if an equal mass of helium (**He**) gas was used instead of hydrogen gas? (27)

**5.** (*a*) (*i*) What is an electric current?

(*ii*) What is meant by the potential difference between two points in an electrical field? (9)

(*b*) (*i*) State Ohm's law.

(*ii*) **Figure 5** shows a circuit, used to verify Ohm's law for conductor **X**, with three essential components missing.
Copy this circuit and complete it in your answerbook to include the missing components.

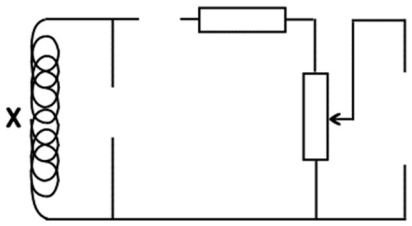



**Figure 5**

(*iii*) In the case of conductor **X** obeying Ohm's law, sketch the graph you would expect to obtain with data provided using the completed circuit. (21)

(*c*) Consider the circuits shown in **Figure 6** and **Figure 7** each of which contains twelve identical 14.4 Ω resistors, a 12.0 V battery and a switch that automatically opens the circuit 600 s after it is switched on. The total resistance of the connecting wires, battery and switch is negligible in each circuit.

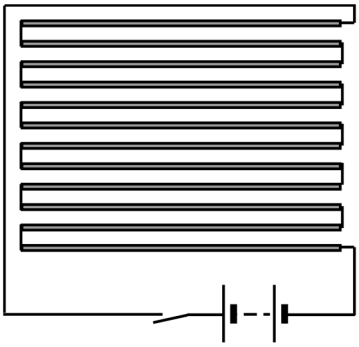


**Figure 6**

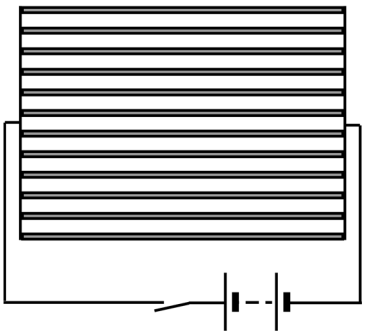


**Figure 7**

In the case of the circuit in **Figure 6**, calculate

(*i*) the total resistance in the circuit,

(*ii*) the current through the resistors when the switch is closed,

(*iii*) the energy converted to heat in *one* of the resistors in 600 s.

In the case of the circuit in **Figure 7**, calculate

(*iv*) the total resistance of this arrangement of resistors,

(*v*) the current through one of the resistors when the switch is closed,

(*vi*) the energy converted to heat in *one* of the resistors in 600 s. (30)

(*d*) The circuits in (*c*) could be used as vehicle rear-window demisters to remove condensation from the inside surface of the window or to melt frost on the outside. Which circuit do you think would be more suitable for use in this way?
Give a reason for your answer. (6)

**6.** Answer any **two** of the following parts (*a*), (*b*), (*c*), (*d*). Each part carries 33 marks.

(*a*) In an experiment to verify the principle of conservation of momentum, **A**, of mass 180 g, was set in motion with constant velocity and was allowed to collide with stationary **B** of mass 198 g. After the collision **A** and **B** moved together with constant velocity. **A** travelled 42 cm in 0.35 s before colliding with **B**. Together **A** and **B** travelled 20 cm in 0.35 s after the collision in the same direction as **A** was originally moving.

(*i*) Draw a labelled diagram of an arrangement of apparatus suitable for this investigation.

(*ii*) State the principle of conservation of momentum. (12)

(*iii*) Calculate the velocity of **A** before the collision.

(*iv*) Calculate the momentum of **A** and **B** travelling together after the collision.

(*v*) Explain whether the data verifies the principle of conservation of momentum. (21)

(*b*) **Figure 8** shows a beam of monochromatic visible light striking a pair of narrow slits. After passing through the slits the light then strikes a screen.

(*i*) Name the two phenomena that occur when the light passes through the slits.

(*ii*) What do these phenomena tell us about the nature of light? (12)

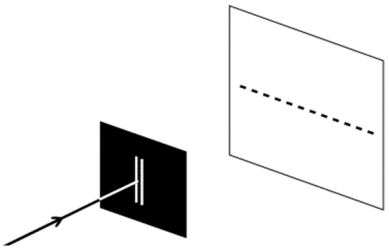


**Figure 8**

When electromagnetic radiation of a certain frequency falls on a freshly sanded zinc plate placed on the cap of a negatively charged gold leaf electroscope, the leaves collapse.

(*iii*) What phenomenon occurs when the radiation strikes the zinc plate?

(*iv*) Explain why visible light does *not* cause the leaves to collapse but ultraviolet light does.

(*v*) Calculate the energy of a photon of electromagnetic radiation from a source that has a wavelength of $1.5 \times 10^{-7}$ m. (21)

(*c*) (*i*) What is meant by electromagnetic induction?

(*ii*) Describe a simple demonstration of electromagnetic induction.

(*iii*) Describe with the aid of a labelled diagram how a step-down transformer works. (21)

A transformer with 250 turns in its primary coil converts 230 V a.c. to 46 V a.c.

(*iv*) How many turns are there in the secondary coil of this transformer?

(*v*) If the current in the secondary coil is 12.5 A, find the current in the primary coil assuming no power loss. (12)

(*d*) An ionising smoke alarm uses a small quantity of a certain radioactive isotope as a source of alpha particles.

(*i*) What is radioactivity?

(*ii*) In terms of subatomic particles, what is an alpha particle?

(*iii*) Identify the isotope used in the smoke alarm that produces neptunium–237 nuclei by alpha decay. (15)

(*iv*) Suggest why an alpha-particle source is more suitable for use in a domestic smoke alarm than either a beta-particle source or a gamma-ray source.

(*v*) Define the half-life of a radioactive isotope.

(*vi*) 6.25% of the radioactive isotope used in a smoke alarm will remain undecayed 1,728 years after its manufacture. What is the half-life of the radioactive isotope? (18)

## SECTION II – CHEMISTRY

**7.** Answer **eleven** of the following items (*a*), (*b*), (*c*), etc. All the items carry equal marks. *Keep your answers short*.

(*a*) How many (*i*) neutrons, (*ii*) electrons, does a dipositive ion of a calcium–40 atom have?

(*b*) Define relative atomic mass.

(*c*) What is the atomic number of an atom that has 10 neutrons and a mass number of 19? Is this atom an isotope of the element oxygen?

(*d*) How many lone pairs of electrons are present in the valence shell of the central atom in a molecule of (*i*) $\mathbf{H_2O}$ (*ii*) $\mathbf{NH_3}$?

(*e*) An instant coffee contains $5 \times 10^{-4}$ moles of caffeine ($\mathbf{C_8H_{10}N_4O_2}$) per spoonful of granules. See **Figure 9**. How many nitrogen atoms are there in the caffeine in one spoonful of these coffee granules?



**Figure 9**

(*f*) What are allotropes? Give an example.

(*g*) A $\mathbf{H_2S}$ molecule has an overall dipole moment but a $\mathbf{BeH_2}$ molecule does not. Account for this difference.

(*h*) Write the chemical formulae for copper(I) oxide and iron(II) sulfate.

(*i*) Write a balanced equation for the reaction between sodium hydride and water.

(*j*) Identify (*i*) an acidic oxide of sulfur, (*ii*) a basic compound of hydrogen and a non-metallic element.

(*k*) Hydrogen and oxygen gases are produced by the electrolysis of acidified water in a voltameter like that shown in **Figure 10**. What is the ratio of $\mathbf{H_2}$ to $\mathbf{O_2}$ produced (*i*) by volume, (*ii*) by mass?

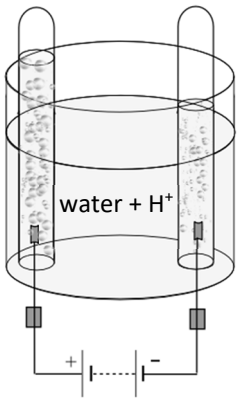



**Figure 10**

(*l*) A laboratory technician diluted 25 $cm^3$ of a 0.014 M solution to 35 $cm^3$ by adding deionised water. What was the new concentration of the solution in moles per litre?

(*m*) Distinguish between an endothermic and an exothermic reaction.

(*n*) Define heat of formation.

(*o*) **Figure 11** shows the structure of 2-anisaldehyde.

(*i*) Copy the diagram into your answerbook and circle the aldehyde functional group.

(*ii*) How many hydrogen atoms has this molecule?

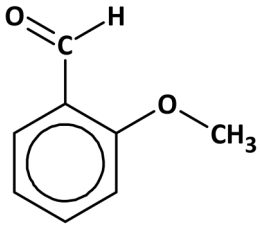



**Figure 11**

(*p*) Give the systematic IUPAC name for an alcohol that has the same molecular formula as diethyl ether ($\mathbf{C_2H_5OC_2H_5}$).

(*q*) Explain why ethanoic acid is very soluble in water.

(*r*) What would you expect to observe, if anything, when a few drops of dilute acidified $\mathbf{KMnO_4}$ solution are added to a little propanone in a test tube? Explain.

(11 × 6)

**8.** The arrangement of the elements in the modern periodic table is consistent with our understanding of atomic structure and how electrons are arranged in the electron cloud of an atom.

(*a*) (*i*) What is an element?

(*ii*) How are the elements arranged in the modern periodic table? (9)

(*b*) (*i*) In terms of arrangement of electrons, what do elements in the same period (row) in the periodic table have in common?

(*ii*) Which of the first 36 elements have their two outermost electrons in an *s* sublevel?

(*iii*) Define an atomic orbital.

(*iv*) Write the *s*, *p*, *d* electron configuration for zinc.
Why is zinc *not* considered to be a transition element?

(*v*) Identify the element, of the first 36 elements, that has the largest atomic radius.
Justify your selection. (33)

(*c*) **Figure 12** represents the first six energy levels in the electron cloud of a hydrogen atom.

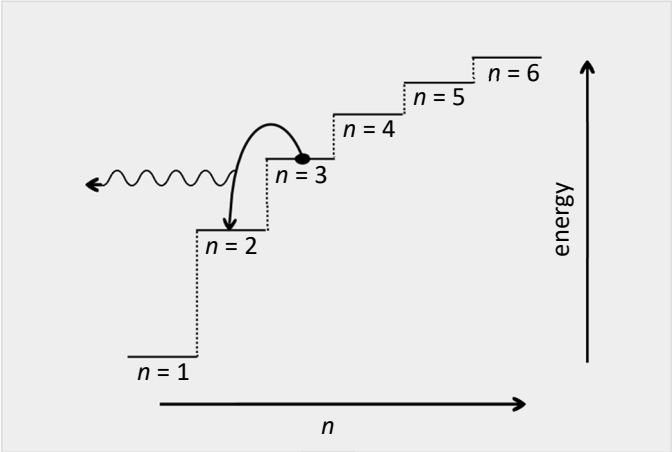



**Figure 12**

(*i*) Name the scientist who first said that an electron in an atom could occupy only certain energy levels.

(*ii*) Explain how $E_3 - E_2 = hf$ applies to the electron in a hydrogen atom moving from the third to the second energy level. (9)

(*d*) The first ionisation energy of sodium is represented by the following equation:

$$\mathbf{Na}_{(g)} \rightarrow \mathbf{Na}^{+}_{(g)} + \mathbf{e}^{-}$$

(*i*) Define the first ionisation energy of a neutral gaseous atom in its ground state.

(*ii*) What is the shape of the atomic orbital from which the electron in the equation above is removed?

(*iii*) Write a set of quantum numbers that describes this electron before it was removed from the sodium atom. (15)

**9.** (*a*) Define, according to Brønsted-Lowry theory, (*i*) an acid, (*ii*) a base.
Identify (*iii*) the conjugate acid, (*iv*) the conjugate base, of $\mathbf{HBO_3^{2-}}$. (12)

Bread soda is a common household chemical. It is a white powder composed of the weak base **$NaHCO_3$** and small quantities of other non-basic chemicals.
To determine the percentage of **$NaHCO_3$** in a bread soda sample, an aqueous solution of the bread soda was first prepared, and portions of this solution were titrated with a standard solution of sulfuric acid. The sulfuric acid was added to a number of 25.0 $cm^3$ portions of the **$NaHCO_3$** solution in a conical flask to which a few drops of indicator had been added.
On average, 16.5 $cm^3$ of 0.05 M sulfuric acid solution were required for neutralisation.
The balanced equation for the titration reaction is:

$$\mathbf{H_2SO_4 + 2NaHCO_3 \rightarrow Na_2SO_4 + 2H_2O + 2CO_2}$$

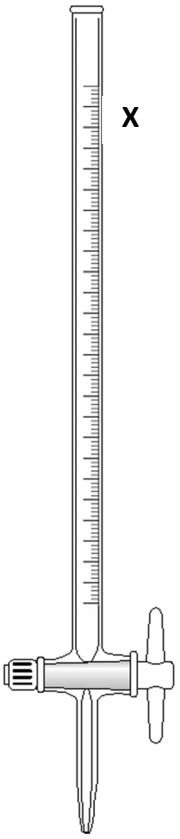


**Figure 13**

(*b*) (*i*) Name the piece of apparatus **X** shown in **Figure 13** which is used to add the sulfuric acid to the conical flask.
Describe the correct procedure
(*ii*) for rinsing **X** before filling it,
(*iii*) for filling **X** with **$H_2SO_4$** for use in the titration,
(*iv*) for taking a reading from **X**. (21)

(*c*) A wash-bottle containing deionised water was used to wash down the sides of the conical flask when approaching the end point of the titration.
Explain why this improved the accuracy of the titration. (6)

(*d*) (*i*) Name a suitable indicator for use in the titration.
(*ii*) What colour change would be observed at the end point using this indicator? (9)

(*e*) Calculate the concentration of **$NaHCO_3$** in the solution titrated
(*i*) in moles per litre,
(*ii*) in grams per litre.
The bread soda solution used in the titrations contained 5.67 grams of powder per litre.
(*iii*) What was the percentage purity of the bread soda powder in terms of **$NaHCO_3$**? (18)

**10.** (*a*) (*i*) In terms of electron transfer, what is a reducing agent?

(*ii*) Identify the reducing agent in each of the following reactions:

$$2Li + 3N_2 \rightarrow 2LiN_3$$

$$2I^- + Cl_2 \rightarrow 2Cl^- + I_2$$ (15)




**Figure 14**

(*b*) (*i*) Arrange the elements **K**, **Cu**, **Zn** and **Fe** in order of their *increasing* ability to act as reducing agents.

(*ii*) Which one of these metals sometimes occurs free in nature?

(*iii*) Identify a metal frequently used as a sacrificial anode to protect iron.
Give two reasons for its suitability for this use. (18)

(*c*) A coin battery like that shown in **Figure 14** contains 0.11 g of lithium usually obtained by the electrolysis of **LiCl**. **Figure 15** shows the electrolysis of molten **LiCl** using inert electrodes. Some **KCl** was mixed with the **LiCl** to lower the melting point of the electrolyte but only the **LiCl** was electrolysed.

(*i*) Explain the underlined terms.

(*ii*) State Faraday's first law of electrolysis.

(*iii*) Write balanced equations for the reactions that occur at the cathode and the anode during this electrolysis.

(*iv*) How long does it take to produce 0.11 g of lithium metal by electrolysis of **LiCl** using a current of 0.50 A? (33)

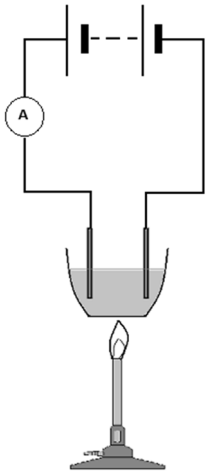



**Figure 15**

**11.** (*a*) (*i*) What is a hydrocarbon?

(*ii*) Name a series of hydrocarbons represented by the general formula $C_nH_{2n}$.

(*iii*) What is the minimum value of ***n*** in this homologous series? (12)

(*b*) Some unsaturated hydrocarbons take part in addition reactions.

(*i*) Explain the underlined term.

(*ii*) Draw the structure of the product of the addition of bromine to ethene.

(*iii*) What observation is made as ethene is bubbled through a bromine solution?

(*iv*) Write a balanced equation for the dehydration reaction of ethanol used to produce the unsaturated hydrocarbon ethene. (21)

(*c*) When methane is mixed with excess chlorine in the presence of ultraviolet light, methane's hydrogen atoms are replaced, one by one, to give a series of four chloroalkanes, the first two of which are chloromethane and dichloromethane.

(*i*) Write a balanced equation for the first reaction in which chloromethane is formed.

(*ii*) Identify the third and fourth chloroalkanes formed in this series of reactions.

(*iii*) Name the organic reaction type involved in the formation of each of these chloroalkanes. (18)

(*d*) (*i*) Identify the two compounds that react together to form ethyl ethanoate.

(*ii*) What catalyst is required for this reaction?

(*iii*) What type of reaction is involved? (15)

**12.** Answer any **three** of the following parts (*a*), (*b*), (*c*), (*d*). Each part carries 22 marks.

(*a*) **Figure 16** shows part of the Guggenheim Museum in Bilbao. The building is clad with thin sheets of an alloy of titanium metal. The metal was extracted industrially from **$TiFeO_3$** ore using reactions described by the following balanced equations.

**$2TiFeO_3 + 7Cl_2 + 6C \rightarrow 2TiCl_4 + 2FeCl_3 + 6CO$**

**$TiCl_4 + 2Mg \rightarrow Ti + 2MgCl_2$**

(*i*) How many moles of **Ti** are there in 6,000 g of the cladding, taking it to contain 92.5% titanium metal by mass?

To produce this quantity of titanium using the reactions above, and assuming they go to completion,

(*ii*) what mass of magnesium would be required,

(*iii*) what volume of **$Cl_2$** gas measured at s.t.p. would be used up,

(*iv*) what mass of **$TiFeO_3$** would be required?



**Figure 16**

(*b*) Define heat of reaction.

In the fertiliser industry ammonia gas is converted to nitrogen(II) oxide gas in the presence of a catalyst according to the following balanced equation.

$$4NH_{3\,(g)} + 5O_{2\,(g)} \rightarrow 4NO_{(g)} + 6H_2O_{(g)}$$

Use Hess's law and the information given below to calculate the heat change for the reaction above.

$$\tfrac{1}{2}N_{2\,(g)} + 1\tfrac{1}{2}H_{2\,(g)} \rightarrow NH_{3\,(g)} \qquad \Delta H = -45.9 \text{ kJ}$$

$$H_{2\,(g)} + \tfrac{1}{2}O_{2\,(g)} \rightarrow H_2O_{(g)} \qquad \Delta H = -241.8 \text{ kJ}$$

$$\tfrac{1}{2}N_{2\,(g)} + \tfrac{1}{2}O_{2\,(g)} \rightarrow NO_{(g)} \qquad \Delta H = 90.3 \text{ kJ}$$

What is the role of a catalyst in a reaction?

(*c*) Define electronegativity.

Distinguish between ionic bonding and covalent bonding.

Explain with reference to bonding

(*i*) the high melting point of **MgO**,

(*ii*) the low boiling point of **$CH_4$**,

(*iii*) why diamond is insoluble in water.

(*d*) Define pH.

Calculate the pH of a 0.5 M **HCl** solution.

**Figure 17** shows the changes in pH as 30 $cm^3$ of 0.5 M **NaOH** solution were added to 20 $cm^3$ of a 0.5 M solution of monobasic acid **A**.

Use **Figure 17** to find

(*i*) the initial pH of the 0.5 M solution of **A**,

(*ii*) the volume of **NaOH** added when the pH was 9.0.

Is acid **A** strong or weak? Explain your reasoning.

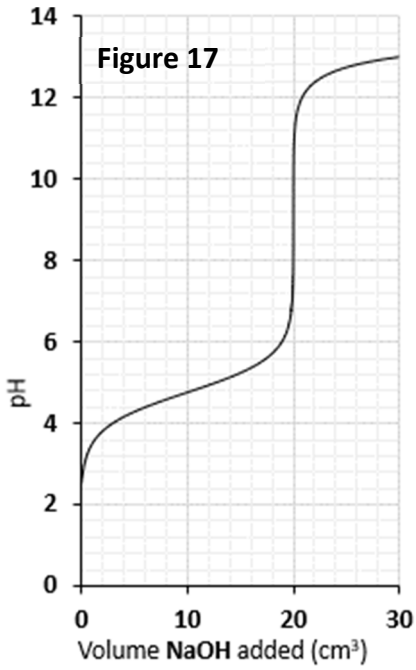



**Figure 17**

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Image Q1(*b*) on page 2: from www.nasa.gov
File: https://www.nasa.gov/sites/default/files/styles/full_width_feature/public/thumbnails/image/pia24127-16.jpg, accessed 13 September 2021

Image Q7(*e*) on page 7: from peter photography at www.pixabay.com
File: instant-coffee-g78bd5d3e8_1920.jpg, accessed 29 September 2021

Image Q10(*c*) on page 10: from www.reichelt.com
File: reichelt.com/de/en/button-cells-lithium-c4211.html, accessed 23 September 2021

Image Q12(*a*) on page 11: from www.maxpixel.net
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Leaving Certificate – Higher Level

# Physics and Chemistry

Wednesday, 22 June
Morning, 9:30 – 12:30